

The user story:

BLASTP on database provided by the user

As a life science scientist

I want to run the BLASTP tool using my own custom database provided in FASTA format. As with the default Uniprot database, the query protein sequence can be provided in text box in FASTA format or uploaded as a FASTA file. When I press the BLAST button, the BLASTP should run using my custom database.

So that I can analyze, using the BLASTP output, which proteins are similar to my query sequence and from what organism these proteins are.

The acceptance criteria

After I put in either a protein sequence in FASTA format in the therefore intended text box or upload a FASTA file of a protein sequence and provide a custom database in FASTA format, the BLASTP output should be written to a file in my downloads folder and shown in a new window when I press the BLAST button.

Importantly, it should not use the default Uniprot database when I provide my own database.

For this test i want to blast the protein GFP (green fluoricent protein) often used in life science experiences to visualize protein expression.

Sequence used:

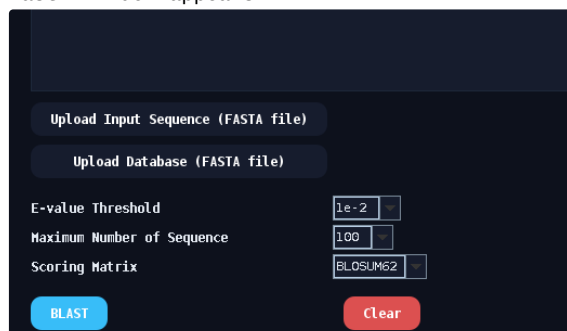
```
>GFP
MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTCLKICTTGKLPVPWPTLVTTFSYGVQCFSRYPDHMKQHDFFKSA
MPEGYVQERTIFFKDDGNYKTRAEVKFEGDTLV
NRIELKGIDFKEDGNILGHKLEYNNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSA
LSKDPNEKRDHMLLEFVTAAGITHGMDELYK
```

`As database the file /BlastImplementation/example\_fasta\_files/protein\_database.txt` was used

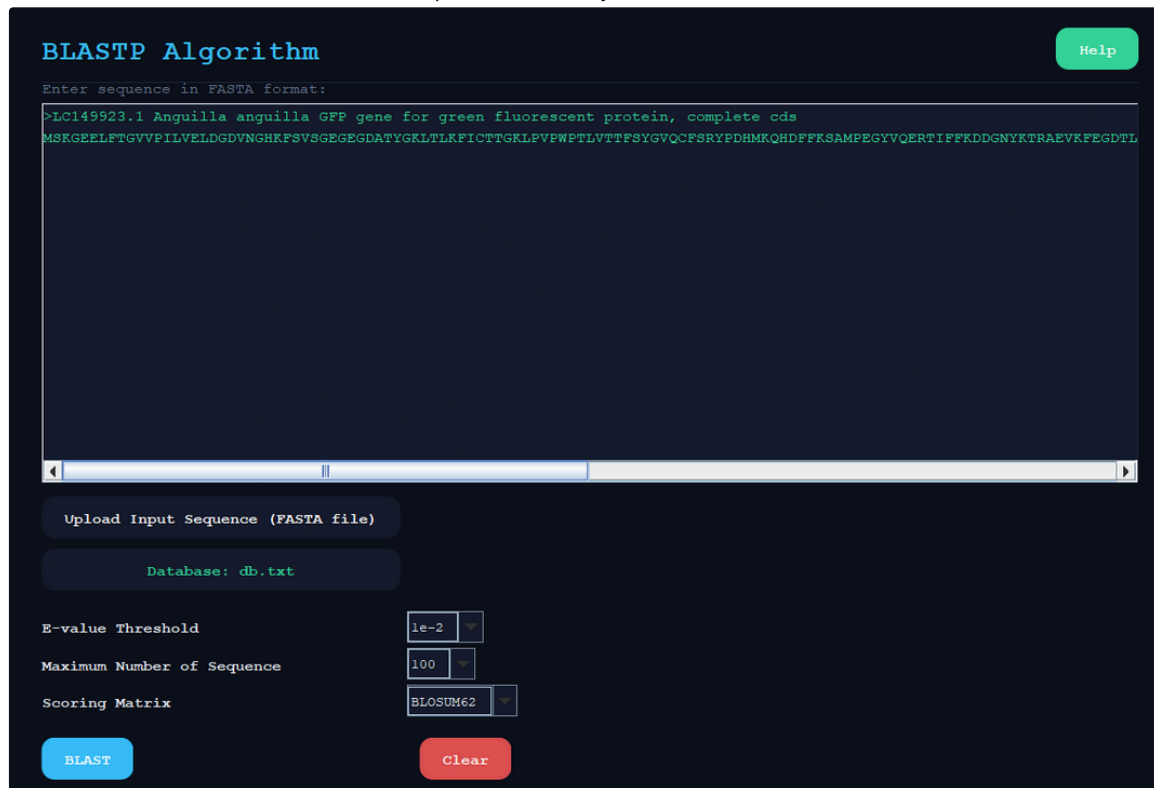
User clicks on BLASTP:



BastP window appears

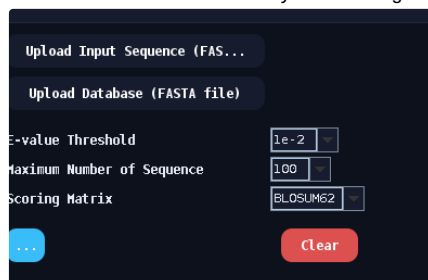


The user can insert the Protein sequence manually in the text box:



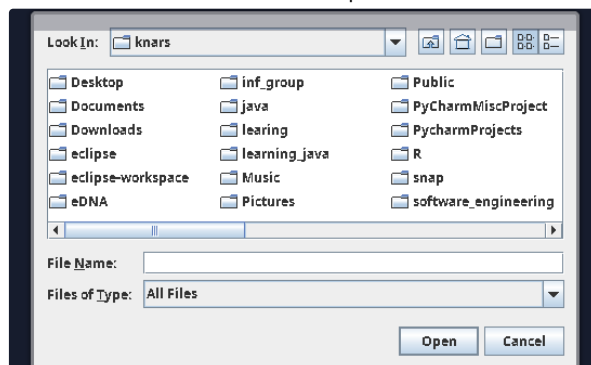
The image shows the BLASTP Algorithm interface. At the top right is a green 'Help' button. Below the title, there is a text box for entering the sequence in FASTA format. The text box contains the following text:
>LC149923.1 Anguilla anguilla GFP gene for green fluorescent protein, complete cds
MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEDATYGKILTKFICTTGKLPVPWPTLVTTFSYGVQCFSRYPDHMKQHDFFKSAMPEGYVQERTIFFKDDGNYKTRAEVKFEGDTL
Below the text box is a scrollbar. Underneath the text box is a button labeled 'Upload Input Sequence (FASTA file)'. Below that is a button labeled 'Database: db.txt'. Below that are three dropdown menus: 'E-value Threshold' set to '1e-2', 'Maximum Number of Sequence' set to '100', and 'Scoring Matrix' set to 'BLOSUM62'. At the bottom are two buttons: 'BLAST' (blue) and 'Clear' (red).

Or select a database by clicking on the upload database button:



The image shows a dialog box for uploading a database. It has a title bar 'Upload Database (FASTA file)'. Below the title bar are three dropdown menus: 'E-value Threshold' set to '1e-2', 'Maximum Number of Sequence' set to '100', and 'Scoring Matrix' set to 'BLOSUM62'. At the bottom are two buttons: '...' (blue) and 'Clear' (red).

A file chooser shows for sequence file section:



The image shows a file chooser dialog. The 'Look in:' dropdown is set to 'knars'. The main area shows a list of folders and files: Desktop, Documents, Downloads, eclipse, eclipse-workspace, eDNA, inf_group, java, learning, learning_java, Music, Pictures, Public, PyCharmMiscProject, PycharmProjects, R, snap, and software_engineering. Below the list is a 'File Name:' text box and a 'Files of Type:' dropdown set to 'All Files'. At the bottom are 'Open' and 'Cancel' buttons.

When the blast button in the blast window is clicked a results window appears with the blast results

BLAST Output

Select Sequence

1

Select Blast Hit

1

ID

reviewed

Description

Green fluorescent protein

Sequence

MSKGEELFTGVVPIILVELDGVNGHKFVSGEGEADATYGKLTLPFICTTGKLPVPMFTLVTFISYGVQCFSRYPDHMQHDFPKSAMPEGTVQERTIFFKDDGNYKTRAEVKFEGDTLVNRIELAGIDFKEDGNIILGHKLE
|||||
MSKGEELFTGVVPIILVELDGVNGHKFVSGEGEADATYGKLTLPFICTTGKLPVPMFTLVTFISYGVQCFSRYPDHMQHDFPKSAMPEGTVQERTIFFKDDGNYKTRAEVKFEGDTLVNRIELAGIDFKEDGNIILGHKLE

Match Sequence

MSKGEELFTGVVPIILVELDGVNGHKFVSGEGEADATYGKLTLPFICTTGKLPVPMFTLVTFISYGVQCFSRYPDHMQHDFPKSAMPEGTVQERTIFFKDDGNYKTRAEVKFEGDTLVNRIELAGIDFKEDGNIILGHKLE

E-value

1.7e-144

Bit Score

497.1

Identity

100.0%

Export Results